

Healthcare Multimodal Foundation Model System

Image + clinical note + labs + structured EHR risk prediction

Project Question

Can multimodal healthcare AI combine medical images, clinical notes, labs, and structured EHR data to improve risk prediction and clinical interpretability?

Overview Metrics **Fairness & Uncertainty** Clinical Interpretation

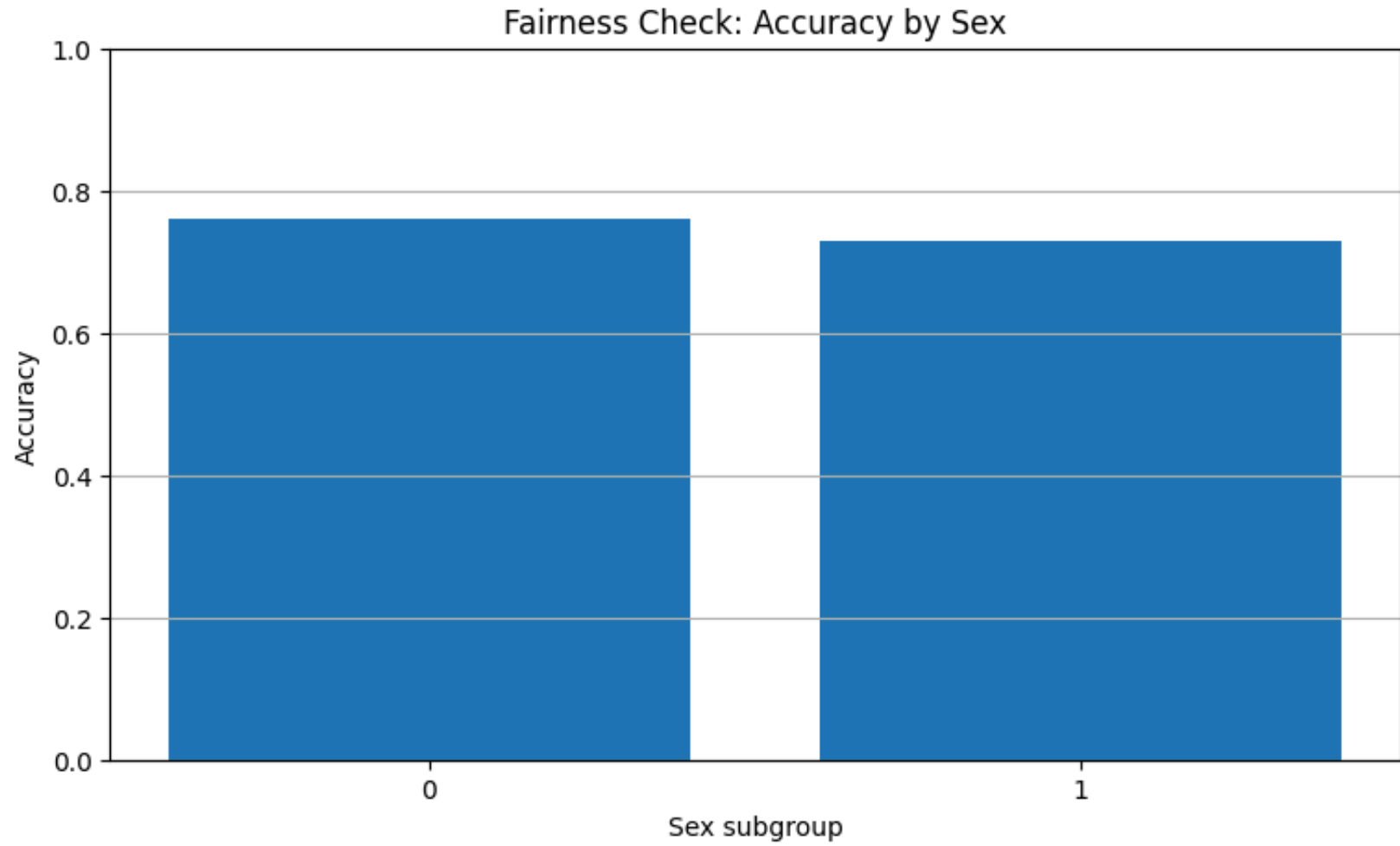
Fairness subgroup metrics

	subgroup_variable	subgroup_value	n	positive_rate	accuracy	f1	auc
0	sex	0	331	0.7613	0.7613	0.8645	1
1	sex	1	294	0.7279	0.7279	0.8425	1
2	site_id	0	150	0.74	0.74	0.8506	1
3	site_id	1	169	0.7633	0.7633	0.8658	1
4	site_id	2	158	0.7532	0.7532	0.8592	1
5	site_id	3	148	0.723	0.723	0.8392	1

High uncertainty review queue

	actual_high_risk	predicted_high_risk	predicted_probability	sex	site_id	uncertainty
15	0	1	0.721	0	2	0.5579
16	0	1	0.7218	1	3	0.5565
17	0	1	0.7218	0	1	0.5563
18	0	1	0.7219	1	3	0.5561
19	0	1	0.7221	0	3	0.5557
20	0	1	0.7223	1	1	0.5555
21	0	1	0.7223	1	1	0.5554
22	0	1	0.7224	0	3	0.5551
23	0	1	0.7228	1	2	0.5545
24	0	1	0.7228	0	2	0.5545





Fairness by sex